

Negative Exponent Law

Teacher's Edition

Read This First

This lesson rebuilds Math Nation 3.4 (SE pp. 118–123) and is the first MA.8.NSO.1.3 lesson of the unit: exponents are integers from here on. The book's discovery is the halving pattern past 2^0 , kept as Notes I; its collaboration tables (product and quotient laws with negative exponents) are Notes III and bank questions 7–8.

The state guide asks for one thing the book does not do explicitly: a task showing the difference between $-b$ and b^{-1} . That is Notes II's contrast line, warm-up question 4, whiteboard 5 and 9, and bank question 9. Expect the -25 answer today; the lesson is built around meeting it.

A note on the book's teacher pages: a thumbnail in the 3.4 differentiation box prints $(-4)^3 \cdot (-4)^{-7}$ as $-1/256$. The student page and the value are $1/256$. Nothing student-facing was wrong.

The book's warm-up is a career-profile video (an ichthyologist) with no mathematics in it. Replaced by spaced retrieval.

Standard and Targets

MA.8.NSO.1.3 Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational-number bases, with procedural fluency.

Clarification Appendix E lists the laws; instruction derives each from expansion. Negative integer exponents yes; fractional exponents no.

Boundary Bases nonzero wherever a negative exponent appears. Rational bases arrive in 3.05; today's bases are integers.

Learning target I can rewrite a power with a negative exponent using a positive exponent, and evaluate it.

Essential question How do you evaluate a numerical expression that has a negative exponent?

Building on MA.7.NSO.1.1 (3.01–3.03) — the seven laws with whole-number exponents; Grade 6 — reciprocals and dividing by a fraction.

Working toward 3.05 — negative exponents on rational bases; 3.06–3.07 — several laws at once; MA.912.NSO.1.1 — rational exponents in Algebra 1.

Where this sits in the course First of four MA.8.NSO.1.3 lessons (3.04–3.07). The Grade 8 FAST reports this under Number Sense and Operations, and the negative-exponent misconception is the one the guide lists first.

Lesson at a Glance

Windy Hill periods run **53 minutes**. The whiteboard round takes the remainder after the taught blocks — **19 minutes** in this lesson — and IXL takes the last 5. Slide numbers below are read from the deck when this edition is built.

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2-3	5 min
Notes	4-6	11 min
Example 1	7-8	3 min
Your Turn	9-10	3 min
Example 2	11-12	3 min
Your Turn	13-14	3 min
Whiteboards	15-32	19 min
IXL	33	5 min
Total		53 min

Materials Whiteboards and markers. Calculators are allowed on everything; on a calculator 5^{-2} returns 0.04, which is $1/25$ and worth showing once.

Where the lesson lives Notes II — 'reciprocal, not opposite'. Whiteboard 5 is the diagnostic that says whether it landed; whiteboard 9 asks for it in writing.

No partner or group work anywhere in this lesson. Every Math Nation exploration this lesson draws on was written for pairs; here the same tables are worked from the front and on individual boards.

Vocabulary

negative exponent law — $a^{-n} = \frac{1}{a^n}$ and $\frac{1}{a^{-n}} = a^n$, for nonzero a

reciprocal — the number that multiplies a given number to give 1: the reciprocal of 8 is $\frac{1}{8}$

opposite — the number the same distance from 0 on the other side: the opposite of 8 is -8 . A negative exponent asks for a reciprocal, never an opposite.

Warm-Up — spaced retrieval

Four questions, two minutes silent, then the reveal. No reteach longer than one sentence per question.

1. Find the value: $\frac{5^4 \cdot 5^0}{5^2}$ **25**

retrieves: yesterday — 3.03 — two laws in one expression

2. Which is greater, $\sqrt{30}$ or 5.5? **5.5**

retrieves: last week — Unit 2 — comparing a root to a decimal (2.05): $5.5^2 = 30.25 > 30$

3. Solve $-3x=21$ $x = -7$

retrieves: last unit — Unit 1 — one-step equations with a negative coefficient (1.03)

4. What is the reciprocal of $\frac{3}{4}$? What is the opposite of $\frac{3}{4}$? $\frac{4}{3}$ **and** $-\frac{3}{4}$

retrieves: prior grade — the prerequisite this lesson needs — Grade 6 — reciprocals (MA.6.NSO.2): reciprocal and opposite are different words

Slide-by-Slide Teaching Notes

Slide 1. Negative Exponent Law 1 min

- Post the learning target. Say the 'today' line and nothing else yet.

Slide 2. Warm-Up — On your own. Four minutes. No notes. 4 min

- Q4 is the whole lesson in one question: a negative exponent asks for the RECIPROCAL of a power, and the misconception is that it asks for the opposite. A board that swaps the two answers is the board to watch today.

Slide 3. Warm-Up — Answers. 1 min

- Reveal. Ask for the band each question came from only if time allows.

Slide 4. Notes — Copy the table. 4 min

- Each entry is the previous one divided by 2. After $2^0 = 1$ the next division gives $\frac{1}{2}$, then $\frac{1}{4}$, then $\frac{1}{8}$. Students fill the last three cells before the rule is stated. The question to ask: is 2^{-1} negative? (No — dividing 1 by 2 gives a positive half.)
- **OFF THE SLIDE, YOURS TO SAY:** 'Negative exponent, positive value. The sign of the exponent tells you to divide; it says nothing about the sign of the answer.'

Slide 5. Notes 4 min

- State the law both ways: a power with a negative exponent is 1 over the same power with a positive exponent, and a negative exponent in a denominator moves up. Then the contrast that the state guide asks for explicitly: $-b$ and b^{-1} are different numbers. Write 5^{-2} , -5^2 , $(-5)^2$ and -25 on the board and ask which are equal.
- **OFF THE SLIDE, YOURS TO SAY:** 'Reciprocal, not opposite. 5^{-2} is one twenty-fifth. It is a small positive number, not a negative one.'

Slide 6. Notes 3 min

- Two roads again: apply the negative exponent law first and multiply the reciprocal out, or apply the product law directly by adding a negative. Same 216. Then the quotient: $9^{-5} \div 9^3 = 9^{-8}$ — subtracting past zero is fine, and the answer is written with a positive exponent at the end.

Slide 7. Example 1 — Negative exponents top and bottom. 1 min

- SPOKEN ONLY. Two negative exponents, different bases — so no product or quotient law applies. Ask: what does each negative exponent tell you to do?

Slide 8. Example 1 — Worked. 2 min

- The 2^{-3} moves down and becomes 2^3 ; the 9^{-2} moves up and becomes 9^2 . Then evaluate. A student who gets $8/81$ moved nothing and just dropped the signs; one who gets $-81/8$ made the exponent's sign the answer's sign.

Slide 9. Your Turn — Same steps, your numbers. Boards up when done.

2 min

- Catches the sign-dropper ($9/5$) and the sign-flipper ($-5/9$). $5/9$ is right.

Slide 10. Your Turn — Answer. 1 min

- Reveal; name what a wrong board most likely did (see the note above).

Slide 11. Example 2 — A quotient law crossing zero. 1 min

- SPOKEN ONLY. Same base, so the quotient law applies: $0 - (-4)$. Ask what 0 minus negative 4 is before anyone writes.

Slide 12. Example 2 — Worked both ways. 2 min

- Road 1: subtract the exponents, $0 - (-4) = 4$. Road 2: 7^0 is 1, and 1 over 7^{-4} is 7^4 by the law's second form. 2401 both ways. A board with 7^{-4} subtracted the wrong way; a board with $1/2401$ read the law backwards.

Slide 13. Your Turn — Same steps, your numbers. Boards up when done. 2 min

- Catches $-4 - (-6)$ done as -10 (a board with $1/(-8)^{10}$) and a dropped sign on the base (a board with -64 — but $(-8)^2$ is 64). 64 is right.

Slide 14. Your Turn — Answer. 1 min

- Reveal; name what a wrong board most likely did (see the note above).

Slide 15. Whiteboards · Question 1 of 9 19 min for the round

- The law, plain. A board with -8^3 or -512 made the exponent's sign the value's sign — the misconception the guide names first.

Slide 16. Whiteboards · Question 1 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 17. Whiteboards · Question 2 of 9

- Now evaluated. -16 and -8 ($2 \cdot -4$) are the two wrong boards to name.

Slide 18. Whiteboards · Question 2 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 19. Whiteboards · Question 3 of 9

- The second form of the law. A board with $1/100$ applied the law to a power that was already in the denominator — it moved nothing.

Slide 20. Whiteboards · Question 3 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 21. Whiteboards · Question 4 of 9

- The base keeps its sign; the exponent supplies the reciprocal. A board with $1/8$ dropped the base's sign; a board with -8 or 8 ignored the reciprocal.

Slide 22. Whiteboards · Question 4 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 23. Whiteboards · Question 5 of 9

- Diagnostic — the one that matters most today. A board full of B means Notes II did not land; go back to the pattern table and divide 1 by 5 out loud, twice.

Slide 24. Whiteboards · Question 5 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 25. Whiteboards · Question 6 of 9

- Diagnostic. The law is unchanged; the only new thing is adding a negative. B is the board that read 'negative exponent' as 'subtract'.

Slide 26. Whiteboards · Question 6 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 27. Whiteboards · Question 7 of 9

- Exponents that sum to zero. The reciprocal route gives $1/81 \cdot 81 = 1$ too. A board with 0 read 3^0 as 0 — the Lesson 3.02 misconception resurfacing.

Slide 28. Whiteboards · Question 7 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 29. Whiteboards · Question 8 of 9

- Subtracting past zero: $-5 - 3 = -8$, then rewrite. A board with 9^{-2} added; a board with 9^{-8} has the right power but not the positive form that was asked for.

Slide 30. Whiteboards · Question 8 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 31. Whiteboards · Question 9 of 9

- Written. The book's own cool-down item, kept. Full credit: the correct value and the sentence 'a negative exponent means the reciprocal of the power, not a negative number'. A student who writes 'she forgot the parentheses' is thinking of $(-5)^2$, a different expression.

Slide 32. Whiteboards · Question 9 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 33. IXL — Last five minutes. 5 min

- IXL: whatever is not finished is tonight's practice; SmartScore 67.

The Whiteboard Round — what each board tells you

Nine questions, one at a time, each with its reveal. Boards down until the cue, all up together, scan the back row first, question the blank boards before the wrong ones. Question 9 is written work. Every wrong option on a multiple-choice question is a named error; the benchmark it comes from is cited.

#	Question	Answer	What a wrong board says
1	8^{-3}	$\frac{1}{8^3}$	-8^3 or -512 — negative exponent read as a negative value [B1G-M 8.NSO.1.3 misconception: $-b$ vs b^{-1}]; $1/8$ — the exponent was dropped
2	2^{-4}	$\frac{1}{16}$	-16 — negative exponent read as a negative value [B1G-M 8.NSO.1.3]; -8 — multiplied base by exponent
3	$\frac{1}{10^{-2}}$	$10^2 = 100$	$1/100$ or 0.01 — the negative exponent was made positive without moving the power [law applied in only one direction]
4	$(-2)^{-3}$	$-\frac{1}{8}$	$1/8$ — the base's own sign was dropped [8.NSO.1.3: rational bases include negatives]; -8 — the negative exponent was treated as positive
5	Which is the value?	$A - 1/25$	(B) the negative exponent was read as a negative value, $-(5^2)$ [B1G-M 8.NSO.1.3 misconception: $-b$ vs b^{-1}]; (C) multiplied the base by the exponent, $5 \cdot (-2)$ [misconception: exponent treated as a factor]; (D) reciprocal of $5 \cdot 2$ instead of 5^2 [reciprocal right, power wrong]
6	Which expression is equivalent?	$A - 6^3$	(B) subtracted instead of adding, $-2 - 5$ [product law: add the exponents, including a negative one]; (C) multiplied the exponents [confused with power of a power]; (D) multiplied the bases as well [7.NSO.1.1 misconception: operating on the bases]
7	$3^{-4} \cdot 3^4$	1	0 — zero exponent read as 0 [B1G-M 8.NSO.1.3]; 81 or $1/81$ — one factor was ignored
8	$\frac{9^{-5}}{9^3}$	$\frac{1}{9^8}$	9^{-2} or $1/81$ — subtracted $5 - 3$ ignoring the sign [integer subtraction, MA.7 prerequisite]; 9^8 — the minus was dropped at the end

#	Question	Answer	What a wrong board says
9	Ellen says the value of 5^{-2} is -25 . Explain why she is wrong, and show the correct value.	$5^{-2} = 1/5^2 = 1/25$	' $-1/25$ ' — took the reciprocal AND the opposite [B1G-M 8.NSO.1.3 misconception]; 'she forgot parentheses' — confused 5^{-2} with $(-5)^2$

Question Bank — variation and answers

Variation. Questions 1-6 rewrite only, one move each: a plain negative exponent, one in a denominator, a negative base, a power of ten, a coefficient that is not a base, two powers with different bases. 7-8 add one law (product, then quotient) with the negative exponent, and vary the sign of the base and whether the answer lands above or below zero. The pattern breaks at 9, where four expressions that look alike are opposites and reciprocals of one another, and at 12, where two options are NOT $1/64$ for two different reasons. The Additional sheet mirrors each position.

The bank and its Additional sheet are posted, never assigned; they feed the quizzes. Answers below are the same values the keys print — both come from one source.

Question Bank

- $\frac{1}{8^3}$
- $(-4)^5$
- $\frac{1}{(-2)^8}$
- 10^2
- $-5 \cdot \frac{1}{6^4} = -\frac{5}{6^4}$
- $\frac{9^2}{2^3}$
- (a) $(-3)^{-3} = \frac{1}{(-3)^3} = -\frac{1}{27}$ (b) $2^4 = 16$ (c) $9^0 = 1$
- (a) $(-5)^{-2} = \frac{1}{25}$ (b) $2^3 = 8$ (c) $3^{-12} = \frac{1}{3^{12}} = \frac{1}{531,441}$
- (a) -4 ; $1/4$; $-1/4$; $-1/4$ (b) 4^{-1} *is the reciprocal of 4*; -4 *is the opposite of 4*.
- (a) $n = -5$ (b) $n = -3$ (c) $n = -4$
- A A**
- A, B, C, D A, B, C and D**

Question Bank — Additional

- $2^{-5} = \frac{1}{2^5}$ (or $\frac{2^3}{2^8}$)
- $-\frac{7}{2^4}$
- 10^3
- $\frac{1}{(-4)^5}$

5. $\frac{1}{9^6}$
6. $(-3)^7$
7. (a) $(-7)^{-2} = \frac{1}{49}$ (b) $8^0 = 1$ (c) $5^3 = 125$
8. (a) $(-6)^{-3} = -\frac{1}{216}$ (b) $3^{-7} = \frac{1}{2,187}$ (c) $5^2 = 25$
9. (a) **-10; 1/10; -1/10; -1/10** (b) 10^{-1} **is the reciprocal; -10 is the opposite.**
10. (a) $n = -4$ (b) $n = -5$ (c) $n = -3$
11. **A A**
12. **A, B, C, D A, B, C and D**

What the audit against Math Nation found

- Math Nation 3.4 student pages verified: the pattern table, $1/16$, $8^{-3} \rightarrow 1/8^3$, $1/(-4^{-5}) \rightarrow -4^5$, 216, $1/256$, 1, 64, 2401, $1/43,046,721$, $(5^{-3})^2 = 5^{-6}$.
- The book writes $1/(-4^{-5})$ without parentheses on the base; the value is the same either way here, but our items always write $(-4)^{-5}$ so that the base's sign is unambiguous.
- Teacher-page thumbnail (TE p. 226) prints $-64/16384 = -1/256$; the student page (p. 233) correctly has $-64/-16384 = 1/256$.
- The book's 3.4.1 career video is not a warm-up under §9; replaced.
- Practice L4 and Homework L4 keys check; their tables are bank questions 7–8 here.

What changed from the book

- 3.4.1 video warm-up $\rightarrow$ four retrieval questions, with the reciprocal/opposite prerequisite in the prior-grade slot.
- 3.4.2 laws-review table $\rightarrow$ folded into 3.03's Notes I (already copied).
- 3.4.3 partner exploration (the pattern) $\rightarrow$ Notes I, taught from the front.
- 3.4.4 Bring It Together and 3.4.5 Guided Practice $\rightarrow$ Notes II and Example 1.
- 3.4.6 partner collaboration $\rightarrow$ Notes III, Example 2 and the round.
- 3.4.7 Cool Down (Ellen) $\rightarrow$ whiteboard 9, written.